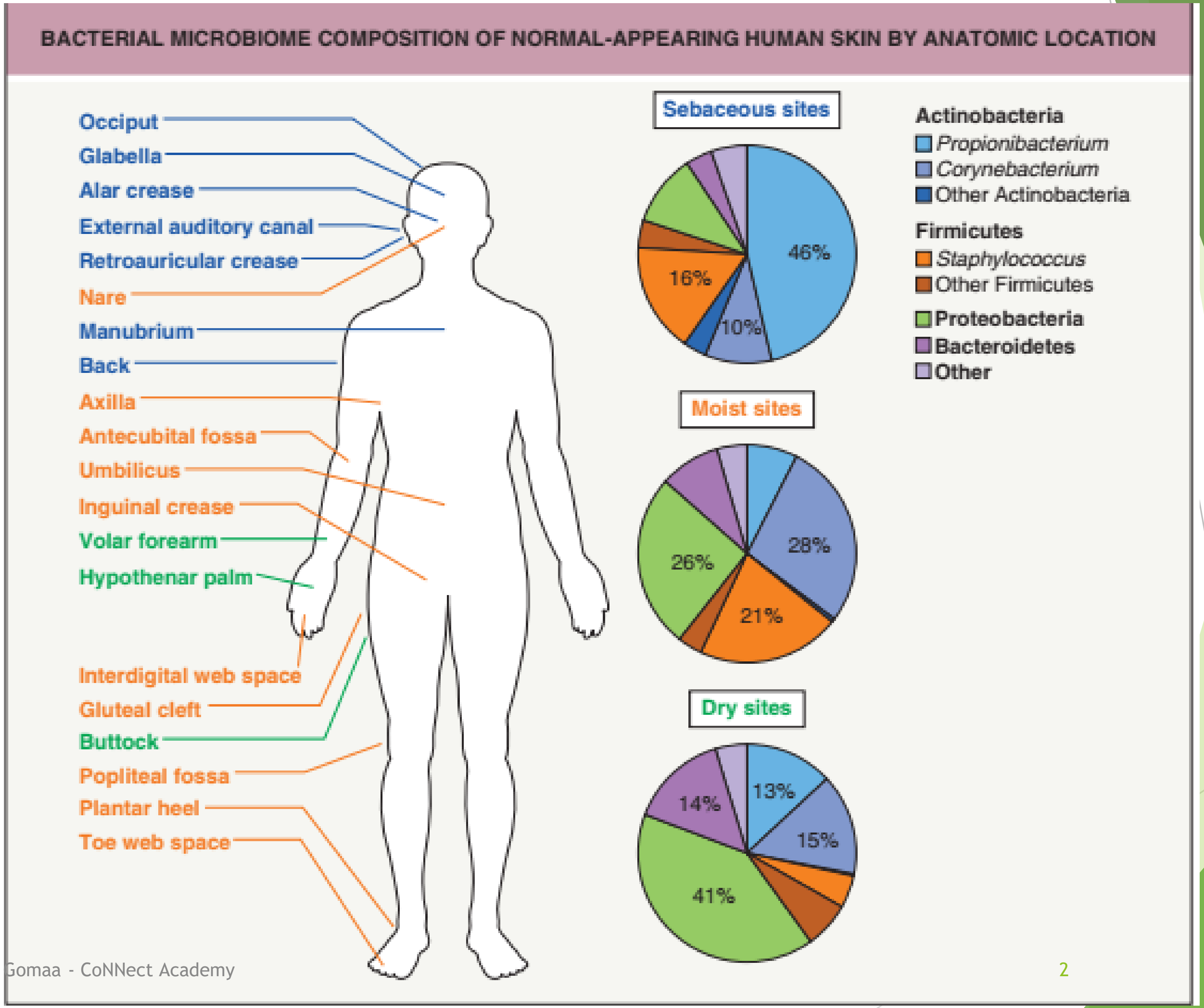
Bacreial infection

Dr eman gomaa

Consultant dermatologist


BACTERIAL MICROBIOME COMPOSITION OF NORMAL-APPEARING HUMAN SKIN BY ANATOMIC LOCATION
Occiput
Glabella
Alar crease
External auditory canal
Retroauricular crease
Nare
Manubrium
Back
Axilla
Antecubital fossa
Umbilicus
Inguinal crease
Volar forearm
Hypothenar palm
Interdigital web space
Gluteal cleft
Buttock
Popliteal fossa
Plantar heel
Toe web space
Sebaceous sites
46%
16%
10%
Moist sites
28%
26%
21%
Dry sites
13%
14%
15%
41%
Actinobacteria
Propionibacterium
Corynebacterium
Other Actinobacteria
Firmicutes
Staphylococcus
Other Firmicutes
Proteobacteria
Bacteroidetes
Other

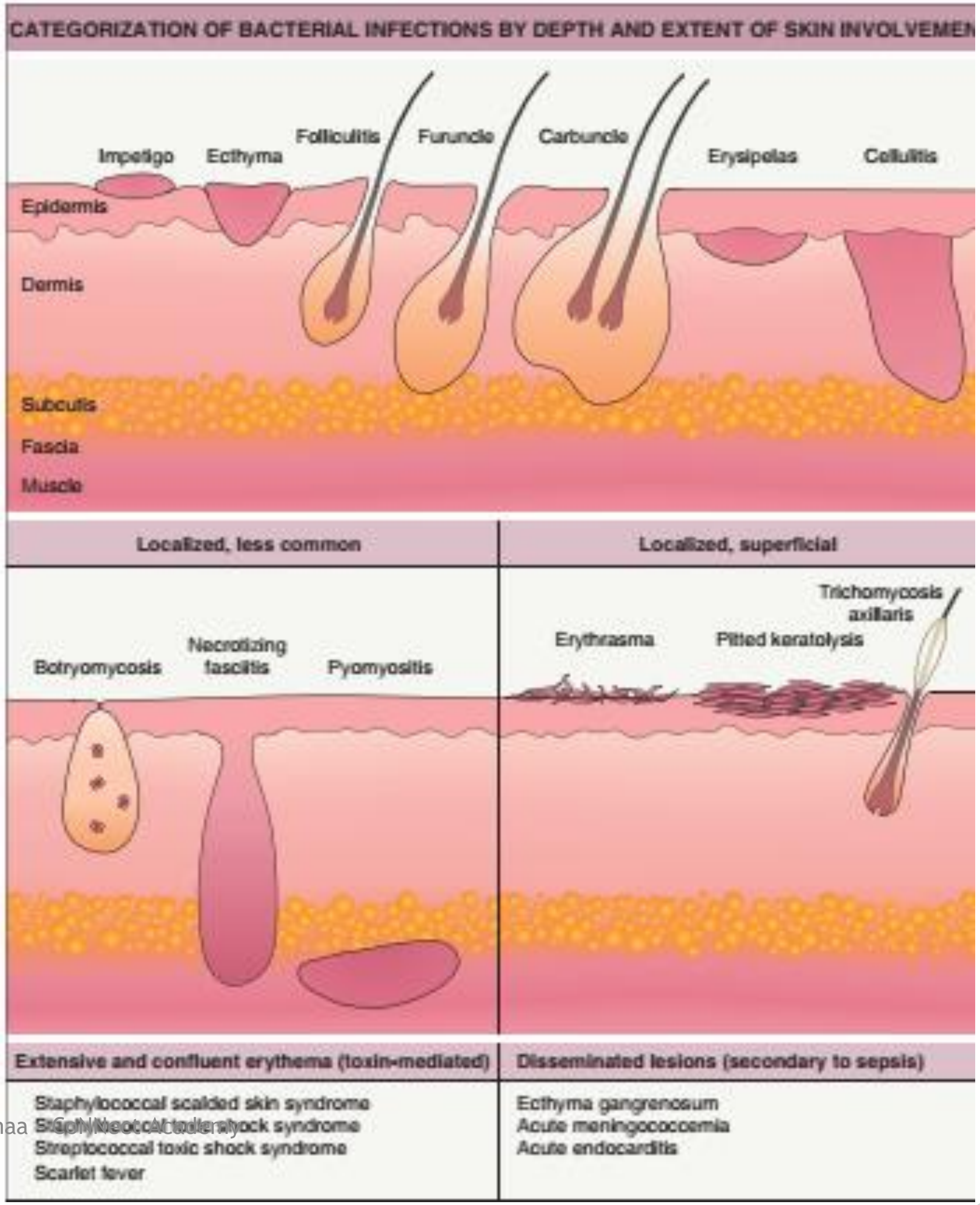
Normal skin flora

Aerobic cocci	s.Aureus s.Epidermids Micrococcus luteus , m roseus	All body Intertregenous area
Aerobic corynibacterium	C. minitissimum C. Lipophilicus C. xerosis	Intertrigenous
Anerobic corynibacterium	Propionibacterium acnes p. Granulisum , avidum	Seb gland , follicles
Gram –ve	Actinobacter spp	Folds
Yeast	Malassezia furfur	Seb gland , scalp

Bacteria make disease in skin

Gram +ve	Gram -ve	spirochetes	Bacteria like fungus	Mycobacteria
• Staph • Strept • Bacilli • Corynibacterium	P .aeruginosum Bartonella	Borrelia Treponema Leptospira	Actinomyces Nocardia	Leprosy T B Non T B

Mechanisim	Staph	Strept infection
Direct infection	Impetigo(bullous) , ecthyma Ocasionally cellulitis Folliculitis .	Impetigo , ecthyma Cellulitis , erysiples Blistering distal dactylitis Necrotizing fasciitis
2ry infection	Eczema , infestation , ulcer	
Toxin	Ssss Scarlatina Tss	scarlet fever Tss
Allergic hypersensitivity		EN , HSP , vasculitis
Skin disease increase by bacteria		Guttat ps , Kawasaki Sweet syndrome Sclersdema adultorum of bushk


CATEGORIZATION OF BACTERIAL INFECTIONS BY DEPTH AND EXTENT OF SKIN INVOLVEMEN
Impetigo
Ecthyma
Folliculitis
Furuncle
Carbuncle
Erysipelas
Cellulitis
Epidermis
Dermis
Subcutis
Fascia
Muscle
Localized, less common
Localized, superficial
Botryomycosis
Necrotizing fasciitis
Pyomyositis
Erythrasma
Pitted keratolysis
Trichomycosis axillaris
Extensive and confluent erythema (toxin-mediated)
Disseminated lesions (secondary to sepsis)
Staphylococcal scalded skin syndrome
Staphylococcal toxic shock syndrome
Streptococcal toxic shock syndrome
Scarlet fever
Ecthyma gangrenosum
Acute meningococcemia
Acute endocarditis

impetigo

- Contageous superficial pyogenic infection
- Caused by staph , strept
- 2 types
  - Non bullous :staph , strept
  - Bollous :staph toxin only
    - Exofoliative toxin →cleavage →Dg1 →superficial blistering
    - If spread to blood →**SSSS**

CHARACTERISTIC FEATURES OF BULLOUS AND NON-BULLOUS IMPETIGO		
	Non-bullous impetigo	**Bullous impetigo**
Epidemiology	• 70% of all cases of impetigo • Children most often affected	• Less common • Often occurs in the neonatal period (see Ch. 34), but children also affected
Clinical lesions	• Early: single 2–4 mm erythematous macule that rapidly evolves into a short-lived vesicle or pustule • Late: superficial erosion with a typical "honey-colored" yellow crust and rapid direct extension of infection to surrounding skin	• Early: small vesicles enlarge into 1–2 cm superficial bullae • Late: flaccid, transparent bullae measuring up to 5 cm in diameter; after rupture there is a collarette of scale, but no thick crust; usually little surrounding erythema
Distribution	• Face (around the nose and mouth) and extremities	• Face, trunk, buttocks, perineum, axillae, and extremities
Associated findings	• Mild lymphadenopathy may be present	• Usually no systemic symptoms but can be associated with weakness, fever, and diarrhea
Clinical course	• Usually a benign, self-limited process • Usually resolves within 2 weeks without scarring if untreated	• Usually resolves in 3–6 weeks without scarring if not treated
Complications	• In 5% of cases, non-bullous impetigo caused by *Str. pyogenes* (M protein types 1, 4, 12, 49, 55, 57, 60) results in acute post-streptococcal glomerulonephritis (APSG)* • Risk of APSG is not altered by treatment with antibiotics • Impetigo has not been linked to a risk of rheumatic fever	• In infants/young children and adults with immunodeficiency or renal failure, exfoliative toxin may disseminate and cause staphylococcal scalded skin syndrome

*Associated with anti-DNase B and antistreptolysin O (ASO) antibodies.

- H p :
  - Non bullous :subcorneal neutrophilis ,spongiosis , exocytosis
    - Late: crus →bacteria, neutrophilis
- Treatment :
  - Disinfatant , cleansing
  - Topical antibiotic
  - Reove crust
  - Sever :systemic , antibiotic
  - Decolonization of nars

DIFFERENTIAL DIAGNOSIS OF NON-BULLOUS AND BULLOUS IMPETIGO		
	Most common	Less common
Non-bullous impetigo	• Insect bites • Eczematous dermatoses • Herpes simplex viral infection • Candidiasis	• Inflammatory tinea corporis/faciei • Varicella • Scabies • Pediculosis • Pemphigus foliaceus
Bullous impetigo	• Bullous insect bite reactions • Thermal burns • Herpes simplex viral infection • Acute contact dermatitis (allergic or irritant)	• Autoimmune bullous dermatoses (e.g. pemphigus foliaceus or vulgaris, linear IgA bullous dermatosis, bullous pemphigoid, dermatitis herpetiformis) • Bullous erythema multiforme • Stevens–Johnson syndrome • Bullous mastocytosis

Ecthyma

- Deep ulcerative non bullus scaring impetigo

CLINICAL FEATURES OF ECTHYMA	
Clinical findings	• Fewer than 10 lesions are typically seen, most commonly on the lower extremities • An initial vesiculopustule enlarges (0.5–3 cm in diameter) over the course of several days, and develops a hemorrhagic crust • The ulcer has a "punched-out" appearance and a purulent, necrotic base • Lesions are slow to heal and produce scarring
Risk factors	• Young age (children), lymphedematous limbs, poor hygiene, neglect (including the elderly), immunosuppression, scratching (e.g. of insect bites), trauma
Complications	• Lesions are often contaminated with staphylococci • Systemic symptoms and bacteremia are rarely seen • Cellulitis and osteomyelitis are extremely infrequent
Diagnosis	• Clinical appearance • Culture of moist, purulent base; skin biopsy with deep-tissue Gram stain and culture occasionally required
Differential diagnosis	• Ecthyma gangrenosum • Ulcers due to vasculitis, vasculopathies, other causes (see Ch. 105)

Fig. 74.5 Ecthyma. Ulceration with hemorrhagic crust on the wrist due to infection with group A streptococci. *Courtesy, Kalman Watsky, MD.*

SSSS
Ritter's disease
Pemphigus neonatorum

- Epidemiology:-infant, young children with decreased renal clearance.
- Age:-Adult with chronic renal insufficiency & immunocompromized.
- Sex:- male predominance 2:1
- Outbreaks in nursuries in neonates.

- ❑ Aetiopathogenesis:- causative:- toxic strains of S.aureus phage group II ET-B.
- ❑ Pathogens:- infection focus → children → nasopharynx & conjunctiva

Adult → pneumonia & bacteremia.

Hematogenous → DSG 1 → superfacial bullae in granular layer split.

Rhinorrhea & conjunctivitis.

Dermo epidermal separation, epidermal nectosis & dermis inflammation & infiltration.

C.Pic
prodroma

- ❑ Malaise, fever & severe tenderness of skin.
- ❑ Erythema head → facial edema → flaccid bullae.
- ❑ +ve Nikolsky → flexure 1st exfoliation.
- ❑ Periorofacil crustation, fissuring & vanish like crusting & no mucosal.
- ❑ .

Fig. 74.6 Staphylococcal scalded skin syndrome. A Diffuse erythema with an early superficial erosion in the antecubital fossa. B More extensive involvement on the neck with a wrinkled appearance of the erythematous skin in addition to peeling and multiple erosions.
Courtesy, Julie V Schaffer, MD.

- H.P.:- subcorneal blistering → no cell in bullae & no organism.
- -ve culture → bullous & blood → ELISA → toxin.
- TTT→ systemic IV AB [hospitalisation].
- Stop drug, IV IG, steroid & supportive ttt.
- Course:- mortality rate →children 3% & adult > 50 %
- with proper ttt resolve in 2 weeks without complications

COMPARISON OF TOXIC EPIDERMAL NECROLYSIS AND STAPHYLOCOCCAL SCALDED SKIN SYNDROME		
	TEN	SSSS
Cause	Usually drug-induced	Infection with toxin-producing *S. aureus*
Age	Primarily adults	Primarily infants and young children
Histology	Dermal–epidermal separation with epidermal necrosis; dermis has a variable inflammatory infiltrate	Granular layer split in epidermis; dermis lacks inflammatory infiltrate
Distribution of rash	Areas of sparing often present	Generalized with flexural accentuation
Mucous membranes	Involved, erosions	Uninvolved
Nikolsky sign	In some areas, difficult to elicit	Present in seemingly uninvolved skin
Face	Vermilion lip erosions and crusting	Perioral and periocular crusting and radial fissuring with mild facial swelling
Treatment	IVIg, cyclosporine, corticosteroids (short course), and TNF inhibitors Supportive care (similar to a burn victim)	Antibiotics (β-lactamase-resistant ± clindamycin) and supportive care

Toxic shock syndrome
acute multi system syndrome

- Causative:- S.aureus mainly streptococci.
- TSST-1 → Superantigen.
- Attached to MHC II without APCS–> massive release cytokine & chemokine.
- Directly activate T cell without APCs. → inflammation → VD & erythema.
- Risk factors :- Tampon → foreign body

No AB for toxin TSST-1

Menstrual TSST-1 → young women.

Non menstrual → both sex → burn, abscesses & post-partum infection

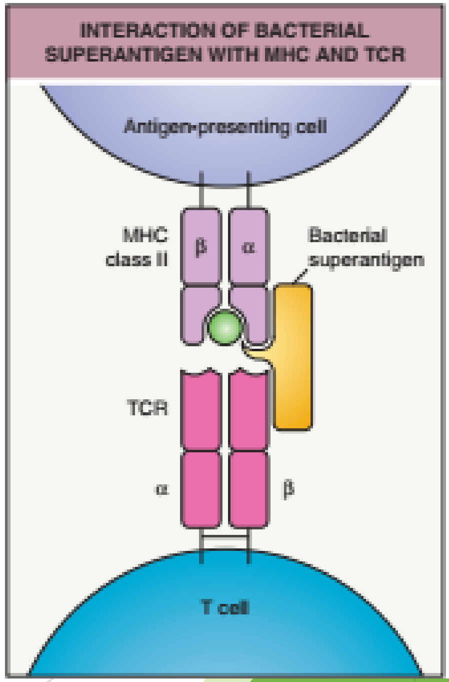


Fig. 74.9 Interaction of bacterial superantigen with the major histocompatibility complex (MHC) and the T-cell receptor (TCR). *Adapted from Janeway CA, Travers P, Walport M, Shlomchik M. Immunobiology: the Immune System in Health and Disease, 6th edn. New York: Garland Science, 2004.*

C/pic

- Fever > 39 degree.
- Desquamation 1-2 weeks after.
- Rash:- diffuse macular erythroderma.
- GIT, mascular, renal & hepatic.

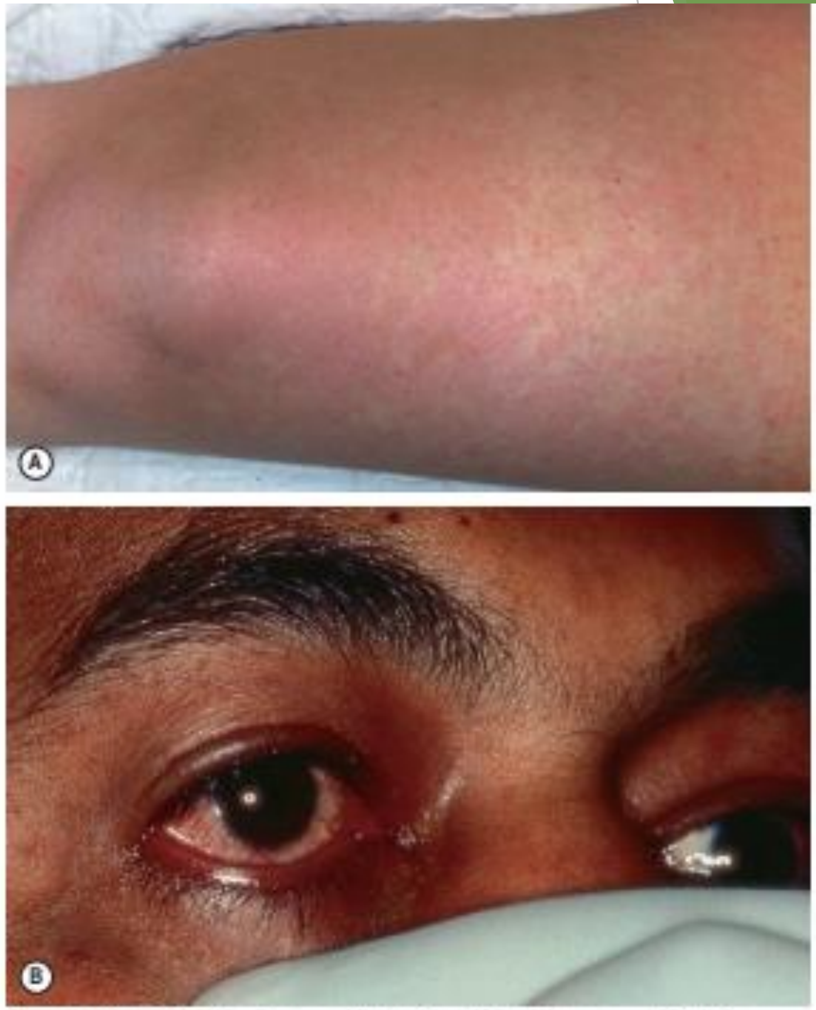


Fig. 74.7 Toxic shock syndrome due to *Staphylococcus aureus* infection. **A** Blotchy erythema is evident on the thigh. **B** Hyperemia of the conjunctiva is seen.

THE TOXIC SHOCK SYNDROMES		
	Staphylococcal	**Streptococcal**
Typical patient	Young (15–35 years) and healthy	Young (20–50 years) and healthy
Diffuse macular erythroderma	Very common	Less common
Vesicles and bullae	Rare	Uncommon (5%)
Localized extremity pain	Rare	Common
Soft tissue infection	Rare	Common
Hypotension	100%	100%
Renal impairment	Common	Common
Predisposing factors	Surgical packing, surgical meshes, abscesses, contraceptive sponge, tampon	Lacerations, bites, bruises, varicella
Positive blood cultures	<15%	>50%
Mortality	<3%	30–60%

Fig. 74.8 Desquamation of the foot following scarlet fever. *Courtesy, Eugene Mirer, MD.*

Scarlet fever scarlatina

- Cauative:- group A streptococci → erythrogenic toxins A,B & C.
- C/pic:- following strept tonsillitis & pharyngitis.
- High fever, sore throat, headache.
- Erythema → neck, chest & axilla.
- Sunburn, with goose pimple.
- Flushed cheeks → circum oral pallor & strawberry tongue.
- Pastia lines:- linear, peticheal → streaks → body folds → inguinal & axilla

- Course:- desquamation of distal digits.
- Complications:- acute glomerulonephritis & rhematic fever.
- DD:- desquamation :- Kwasaki, TSS & viral with high fever.
- Exanthema :- drug, viral, SSSS, JSS & kwasaki.
- TTT:- Penicillin, amoxicillin for 10-14 days.

Erysipelas

St. anything fire

- Group A strept only
- Dermis + upper SC bacteria enter dermis
- Age extremities
- Increases in summer & women > men.
- nities [face]

Cellulitis

- Group A. strept, S.aureus & H.inflenza.
- Deep dermis, SC tissue via break in skin barrier
- Lymphedema, DM, drug recurrent in damage of lymphatic system
- Boys > girls & increases in summer
- Lower extremities

Fever, chills, malaise
hotness, redness, pain & sweating

- Sharply demarcated
- Raised border.
- Less edema (red)
- In severe → bullae,pustules, he, necrosis, lymphangitis

- Illdefined, non-palpaple border
- More edema (pink)
- Same in severe cases

- Erysipeloid, necrotizing fascitis.
- 1-2 weeks with good A.B.
- Pseudo-cellulitis
- Recurrence

- Complications of both:-

Recurrence →LL→ lymphedema.

Rare necrotizing fasciitis → Glomerulonephritis → strept.

Treatment

Mild :oral antibiotic →flucloxacillin 500 mg 4 time /day

ttt of severe forms of both

Benzyl penicillin 600-1200 mg 6 hours IV for 10-14 days – drain abscess & anticoagulant.

Prophylaxis against recurrence → penicillin 250 mg 2 time/d for 6 months.

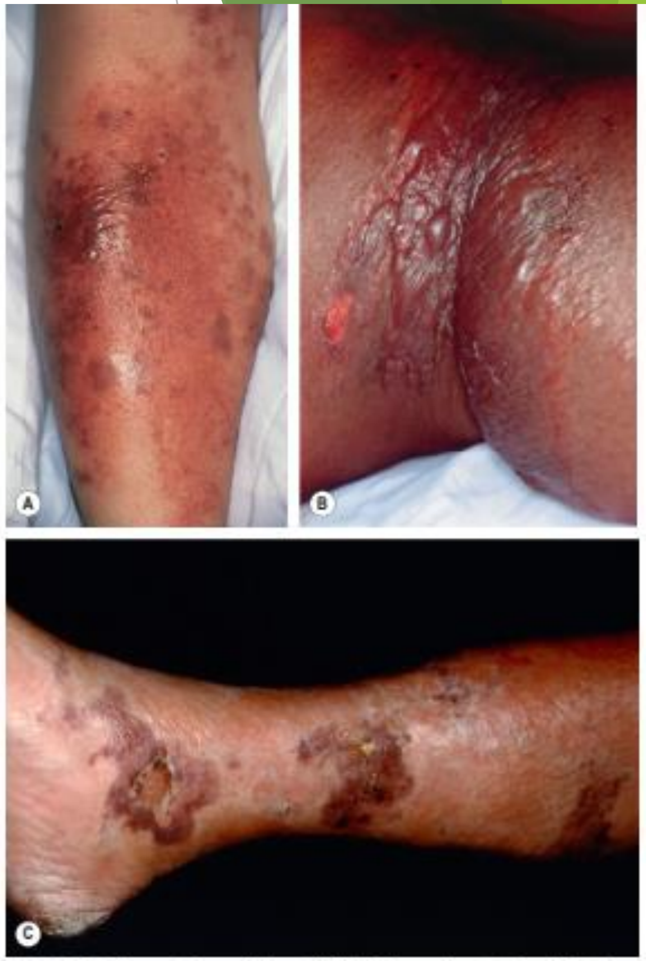


Fig. 74.11 Bullous and necrotizing cellulitis. A Extensive soft tissue infection of the lower extremity due to group A streptococcal infection. B Edema and confluent bulla. C Multiple areas of necrotic crusting and focal purulence due to streptococcal cellulitis.

Pseudocellulitis

1) Sweet's syndrome
2) HZ
3) Well's syndrome
4) Allergic contact dermatitis
5) Angioedema
6) FDE
7) Gout
8) Eryseploid skin metastasis

Eryseploid [Diamond skin disease]

Org:- erysiplothrix rhusopahae

Fissure man & politary

C/pic:- localized :- painful erythematous cellulitis

Finger web sparing distal phalanges

Generalized:- fever- arthralgia-endocarditis

ttt: penicillin

Fig. 74.23 Erysipeloid. Erythema and edema with vesicle formation on the hand and fifth

Bacterial folliculitis

Superficial folliculitis
Bockhart impetigo
S –aureus most common
Pruritic tender small pustules →face scalp , chest
Heal with out scar
Tt:antiseptic , topical antibiotic

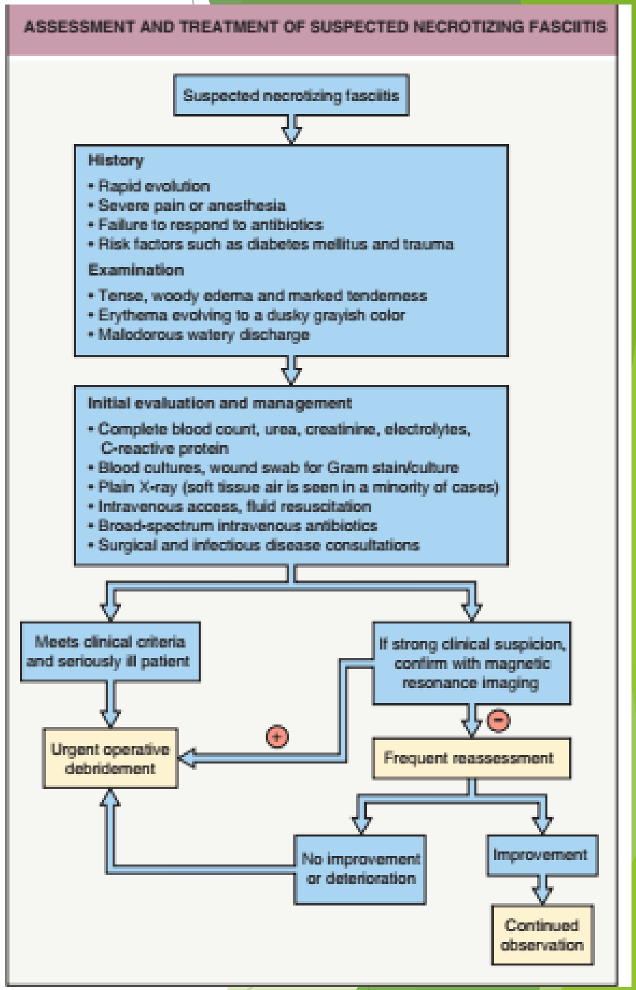
Fruncls , boil
Abscess of hair follicle ,surrounding tissue
Staph aureus , anaerobic bacteria
Pyemia , septecemia
Tt:warm compression
Incusion , drainage
Systemic antibiotic

Carbancle
Collection of fruncles
s. Aureus
Male> female
Tender mass with discharge from follicular orfices
Nape of neck

Other staph , strept infection

1) Necrotizing fasciitis

Rapidly progressive necrosis of sc fat , fascia
P.F :penetrating injury , iv drug use
Organisim :children group A strept
Adult :poly microbial , ecoli, colestridium
Site:extremeties , genitalia , perenium
V painfull erythamatus swelling →gray blue color
→he bullae
Tt:depridment ,
Broad spectrum AB



eFig. 74.6 Necrotizing fasciitis. Necrosis of the subcutaneous fat and fascia of the inner aspect of the upper arm in an elderly patient with diabetes mellitus. Note the watery discharge. *Courtesy, Jean L Bolognia, MD.*

Fig. 74.16 Necrotizing fasciitis. Tense, woody edema of the forearm with purple–gray areas of necrosis and bullae with watery discharge. Intravenous drug use was a predisposing factor. *Courtesy, Luis Requena, MD.*

Other staph & strep infections

Perianal strept dermatitis.
Group A strept
Sharply dermarcated bright red erythema
Perianal pruritis & irritation
Child boy < 4 ys.

Strept intertrigo
Group A strept
Yong child & infant
Erythematous patch with bad odor
Treated by oral penicillin 10 days

Blistering distal dactylitis

Valvar fat bad of finger
Strept

Potriomycosis

Bacterial pseudomycosis

Staph aureus & pseudomonas

SC nodule & ulcer → verrucous plaque → multiple sinuses & fistula → discharge yellow granules.

Extrimities

DD:- Madura foot

TTT sjurgical depridement & systemic AB

Pyomyocitis

In skeletal muscles

S.Aureus

Woody induration in skin

MRI

IV AB

Corynebacteria

- erythrasma
- Trichmycosis axillaris
- Pitted cheratolysis

Erythrasma

- Superficial chronic skin infection
- Due to excessive proliferation of chroynebacterium minutissimum
- C/pic:- well defined pink to brown patch with fine scales in groin & axilla.

  - Variants:-

- Disciform erythlasma :- type II DM.
- Interdigital erythlasma.

Fig. 74.19 Erythrasma. **A** Pink to brown scaly patches on the upper inner thighs. **B** Coral-red fluorescence upon illumination with a Wood's lamp. **C** Hyperpigmented plaques in the inguinal and periumbilical areas. **D** Well-demarcated, scaly, hyperpigmented plaque of disciform erythrasma. *A, B, Courtesy, Louis A Fragola, Jr, MD.*

- Complications :-

    cellulitis & bacteraemia

Investigations:-

    Wood's light → Coral red →porphyrin producing bacteria.

    G +ve bacteria.

    Tissue culture media 199

ttt:-

    Erythromycin & clindamycin

    Oral erythromycin & tetracyclin

Trichomycosis axillaris

- Trichomycosis nodosum
- C/pic:-

  concretions on shaft of the axillary hair.

  Adherent red black nodules.

  Characterisitic odor & red sweat

Inv:-

  Wood’s light → pale yellow

  G+ve

Ttt:-

  Shaving & topical erythromycin

Fig. 74.21 Trichomycosis axillaris. Cylindrical sheaths and beading of the

Pitted keratolysis

- Non inflammatory bacterial infection of the palmo-plantar skin.
- RF:- hyperhidrosis, occlusion & increased PH
- Organism:- kytococcus sedentarius.

  Serine proteas → degradation of keratin

  Sufur containing compounds.

C/pic:- crater like depressions in weight bearing regions

No erethema

DD:- Palmoplantar pits.

Ttt:- topical erythromycin & clindamycin & ttt of hyperhydrosis

Fig. 74.20 Pitted keratolysis of the plantar surface of the foot. Multiple small (A) and larger coalescing (B) craters with decreased stratum corneum that favor pressure points on the plantar surface. *Courtesy, Kalman Watsky, MD.*

Pseudomonas aurgenosa [G –ve]

Green nail syndrome	Pseudomonal bioderma	Pseudomonal blastomycosis like biderma	Malignant otitis externa [swimmer's ear]	Hot tub folliculitis	Hot foot syndrome	Ecthyma gangerinosum
• Frequent water exposure. • Green black nail. • DD:- hematoma, navus & melanoma. • TTT:- topramycin solution, quinolones & aminoglycosides	• Blue-green purulence. • Grape guice like. • Mouth eaten app.  Fig. 74.26 Superficial infection of the skin with *Pseudomonas*. Note the maceration, erosions, and moth-eaten appearance of the skin. Courtesy, Kalman Watsky, MD.	• Large verrucous plaque. • ID. • TTT:- AB & surgical 	• Green purulent discharge. • Tympanic M is not involved.	• Pool with low chlorine level. • Erythema, edema, peri-follicular papule & pustule resolve spon in 1-2 weeks. • Sparing face & neck.	• Painful erythema. • Red to purple nodules. • Pool (self limited)  Fig. 74.27 Pseudomonas hot-foot syndrome. Tender erythematous nodules on the heel. Courtesy, Justin J Green, MD.	• Groin. • Bacteremia & septicaemia. • ID. • Red to purple macule patch → central necrosis → hemorrhagic bullae

Fig. 74.25 Green nail syndrome. Blue–green discoloration
pyocyanin produced by *Pseudomonas aeruginosa*. Note the
onycholysis. *Courtesy, Julie V Schaffer, MD.*

Fig. 74.28 Ecthyma gangrenosum. Embolic lesion of *Pseudomonas aeruginosa* on the chest. Note the necrotic center and inflammatory

Other gram -ve

Partonella
Henselae → cat scratch disease
& basillary angiomatosis.
Basilliformis → partonellosis.
Quintana → trensh fever & basillary angiomatosis

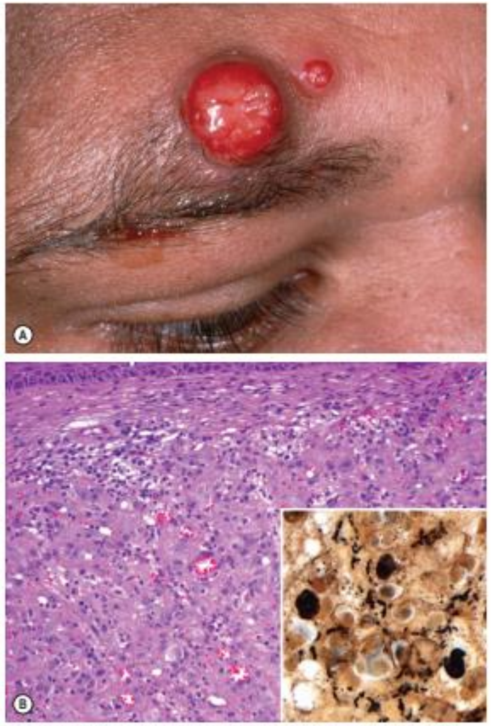
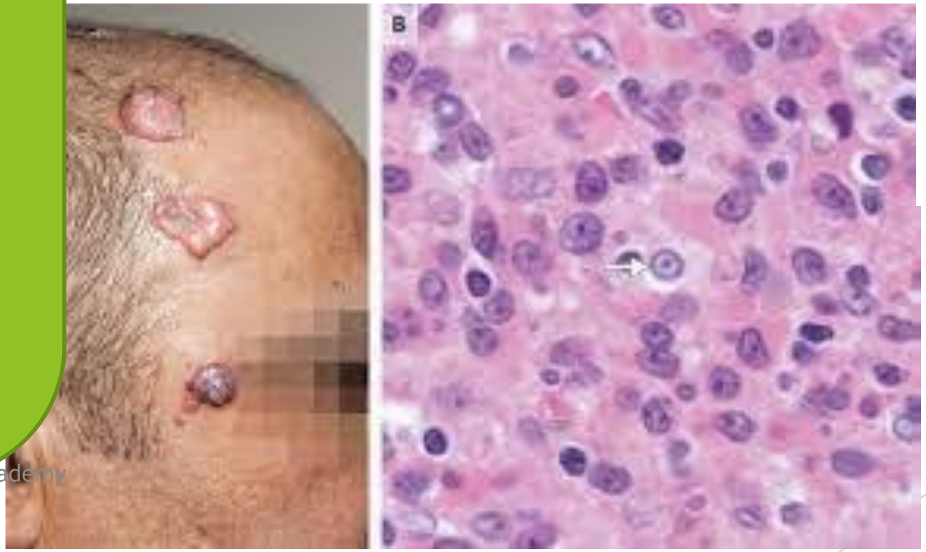
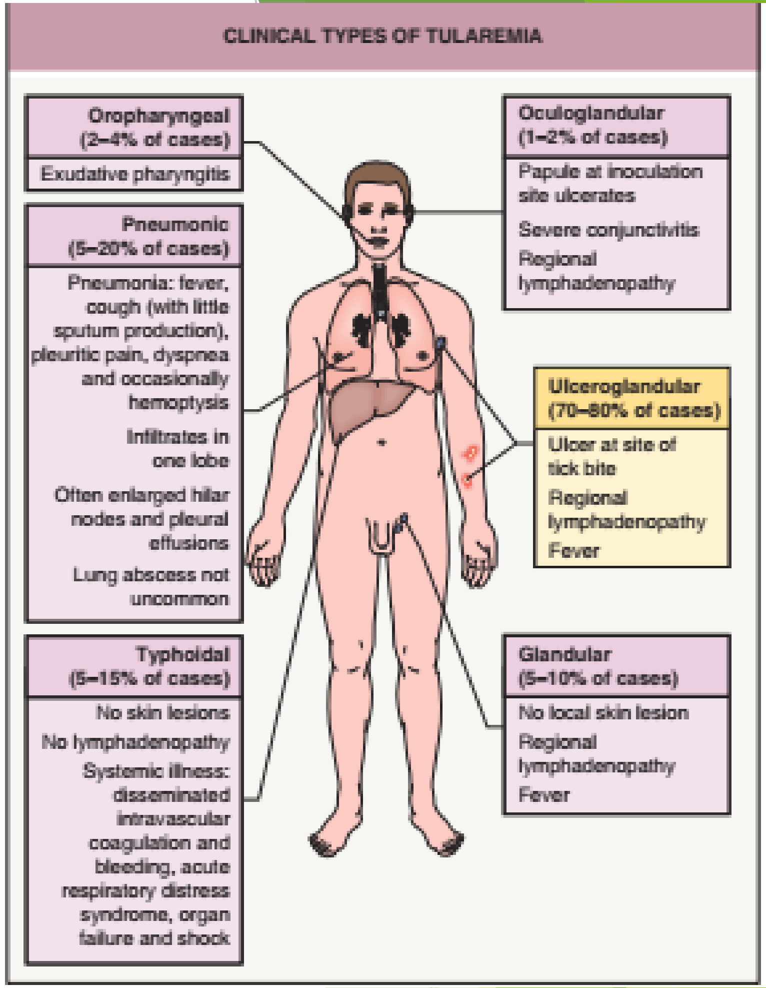


Fig. 74.29 Bacillary angiomatosis. A Bright red nodule and papule on the forehead. **B** Histologically, a dermal proliferation of vessels with plump endothelial cells is evident and scattered neutrophils are present in the accompanying infiltrate. Bacilli (in this case *Bartonella henselae*) are identified with the Warthin–Starry stain (inset). *B, Courtesy, James Patterson, MD.*

Malakoplakia
1)Chronic infection → granulomatous inflammation with large macrophages.
C/pic:- peri anal erythematous indurated nodule → H.P. → micheil gut man body & von hanseman cell
Ttt:- quinolones & clofazimin

Tuleremia [rabbit fever]
Franchicilla tulrenosis
Ttt:- streptomycin 10 days





FOUR MAJOR CLINICAL FORMS OF CUTANEOUS NOCARDIOSIS	
Primary	
Actinomycotic mycetoma	• Half of all cases of actinomycotic mycetoma are due to *Nocardia* species* • Traumatic inoculation causes a painless nodule that enlarges, suppurates, and drains via sinus tracts • Purulent discharge contains sulfur granules • The foot is the usual site of involvement • May involve underlying muscle and bone
Lymphocutaneous	• Occurs days to weeks after trauma • Appears as a crusted pustule or abscess resistan to antibiotics • Ascending lymphatic streaks, a sporotrichoid pattern of papulonodules, and tender palpable lymph nodes may be seen
Superficial cutaneous	• Traumatic implantation of foreign objects (including soil and gravel) into the skin • The diagnosis is based on a high index of suspicion, lack of response to routine antibiotic treatment, and laboratory results
Secondary	
Pulmonary/ systemic	• Subcutaneous abscesses of the chest wall • Pustules, nodules, cutaneous fistulae • Almost universally fatal if left untreated • Most commonly caused by *Nocardia asteroides*

*In Mexico and Central and South America, *N. brasiliensis* is the etiology of 90% of actinomycotic mycetomas, whereas in the US, most mycetomas are caused by true fungi.

actinomycosis

- Ss.c chronic bacterial infection
- Granulomatus inflammation
- Sinus formation
- Actinomyces Israeli
- Cervicofacial
- Pulmonar
- Git

Fig. 74.36 Cervicofacial actinomycosis or "lumpy jaw". Soft tissue swelling and draining erythematous nodules are seen. The discharge contained sulfur granules. *Courtesy, Joyce Rico, MD.*

thankyou.